Lecture 8: The Bootstrap

Resampling · Bootstrap CIs · Permutation Tests

Previously, on Lecture 7...

Sampling distribution: $\hat{\theta}$ is random. Repeat the experiment $\Rightarrow$ different $\hat{\theta}$ each time.

Standard error: $\text{SE} = \text{SD}(\hat{\theta})$. Measures precision of the estimate.

Wald CI: $\hat{\theta} \pm z_{\alpha/2} \cdot \text{SE}$. Relies on asymptotic normality.

t**-interval:** Use when σ is unknown and n is small.

Delta method: CI for $g(\hat{\theta})$ via $\text{SE}(g(\hat{\theta})) \approx |g'(\hat{\theta})| \cdot \text{SE}(\hat{\theta})$.

Today: What if there's **no formula** for the SE?

What if the sampling distribution isn't Normal? What if the statistic is complicated?

The Problem

Lecture 7 world:

Mean, proportion, MLE
SE formulas exist
Normal approx-
imation works

Real world:

Median, correlation, ratio
SE formula is
hard or unknown
Distribution
may be skewed

Bootstrap solution:

Let the *data itself*
tell you the SE
No formulas needed

The Bootstrap Idea

"Pull yourself up by your own bootstraps"

The Core Insight

Ideal world

Draw many samples from the *true* population F

Compute $\hat{\theta}$ on each

$\Rightarrow$ exact sampling distribution

We can't do this

so we do this instead

Bootstrap world

Draw many samples from the *observed data* $\hat{F}_n$

Compute $\hat{\theta}^*$ on each

$\Rightarrow$ approximate sampling dist.

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$\Rightarrow$ approximate sampling dist.

Key idea: The empirical distribution $\hat{F}_n$ is our best estimate of F. Sampling from $\hat{F}_n$ = sampling **with replacement** from the data.

The Bootstrap Algorithm

Step 1: Start with your original data $\mathbf{x} = (x_1, x_2, \ldots, x_n)$.

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↓

Step 5: Use the distribution of $\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*$ to estimate SE, bias, or CI.

Visualizing a Bootstrap Sample

Original data: $\mathbf{x} = (2, 5, 7, 3, 8, 1, 6), \quad n = 7.$

Original:

2	**5**	**7**	**3**	**8**	**1**	**6**

Note: some values appear **multiple times**, others **not at all**. That's "with replacement."

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Boot #1:	**5**	**3**	**3**	**8**	**2**	**5**	**7**

$\hat{\theta}_1^* = \text{median} = 5$

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Boot #1:	**5**	**3**	**3**	**8**	**2**	**5**	**7**	$\hat{\theta}_1^* = \text{median} = 5$
Boot #2:	**7**	**7**	**1**	**6**	**8**	**3**	**8**	$\hat{\theta}_2^* = \text{median} = 7$

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Visualizing a Bootstrap Sample

Original data: $\mathbf{x} = (2, 5, 7, 3, 8, 1, 6), \quad n = 7.$

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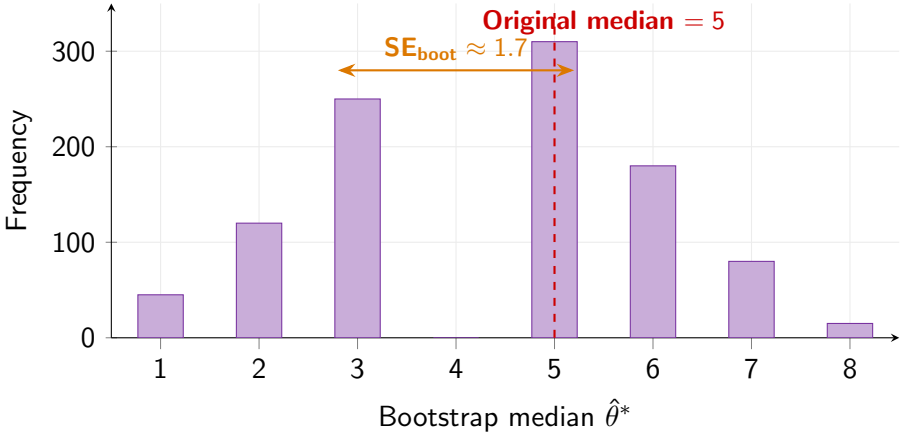
Boot #1:	5	3	3	8	2	5	7	$\hat{\theta}_1^* = \text{median} = 5$
Boot #2:	7	7	1	6	8	3	8	$\hat{\theta}_2^* = \text{median} = 7$
Boot #3:	2	1	6	2	3	1	5	$\hat{\theta}_3^* = \text{median} = 2$

Note: some values appear **multiple times**, others **not at all**. That's "with replacement."

Bootstrap Distribution of the Median



The **spread** of the bootstrap distribution estimates the standard error.
No formula needed — just simulation!

Bootstrap SE & Bias

Estimating uncertainty from the data

Bootstrap Standard Error

$$\widehat{\mathrm{SE}}_{\text{boot}} = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(\hat{\theta}_b^* - \bar{\hat{\theta}}^*\right)^2}$$

where $\bar{\hat{\theta}}^* = \frac{1}{B}\sum_{b=1}^{B}\hat{\theta}_b^*$ is the mean of the bootstrap replicates.

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$B = 200$: rough estimate

$B = 1{,}000$: reliable SE

$B = 10{,}000$: for CIs

More B = more precision in the SE estimate. The cost is purely computational.

Bootstrap Bias Estimation

The bootstrap can also estimate **bias**:

$$\widehat{\text{Bias}}_{\text{boot}} = \bar{\hat{\theta}}^* - \hat{\theta}$$

If the bootstrap mean $\bar{\hat{\theta}}^*$ is systematically shifted from the original $\hat{\theta}$, the estimator is biased.

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$$\hat{\theta}_{\text{corrected}} = \hat{\theta} - \widehat{\text{Bias}}_{\text{boot}} = 2\hat{\theta} - \bar{\hat{\theta}}^*$$

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Caution: Bias correction adds variance. Only use when bias is large relative to SE.

The bias-corrected estimate can have *higher* MSE than the biased one (Lecture 3 tradeoff!).

Why Does the Bootstrap Work?

Real world

$$F \xrightarrow{\text{sample}} \mathbf{x} \xrightarrow{T} \hat{\theta}$$

Variation of $\hat{\theta}$ around θ
= sampling distribution

analogous

Bootstrap world

$$\hat{F}_n \xrightarrow{\text{resample}} \mathbf{x}^* \xrightarrow{T} \hat{\theta}^*$$

Variation of $\hat{\theta}^*$ around $\hat{\theta}$
= bootstrap distribution

Bootstrap principle: The relationship
$\hat{F}_n \to \hat{\theta}^*$ *mimics* the relationship $F \to \hat{\theta}$.
Formally: $\hat{\theta}^* - \hat{\theta} \approx_d \hat{\theta} - \theta$ as $n \to \infty$.

Bootstrap Confidence Intervals

Three methods, from simple to sophisticated

Method 1: The Normal Bootstrap CI

If the bootstrap distribution looks roughly Normal:

$$\hat{\theta} \quad \pm \quad z_{\alpha/2} \cdot \widehat{\mathrm{SE}}_{\text{boot}}$$

Same as the Wald CI from Lecture 7, but with bootstrap SE instead of a formula.

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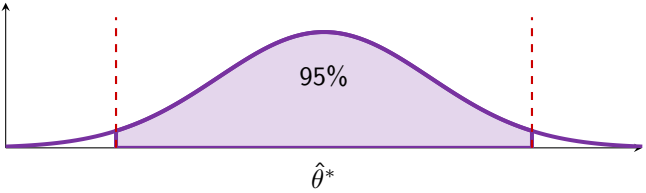

Simple, but assumes the bootstrap distribution is symmetric. Often it isn't!

Method 2: The Percentile Bootstrap CI

$$\text{CI}_{95\%} = \left[\hat{\theta}^*_{(0.025)}, \quad \hat{\theta}^*_{(0.975)}\right]$$

Just use the 2.5th and 97.5th percentiles of the bootstrap distribution.

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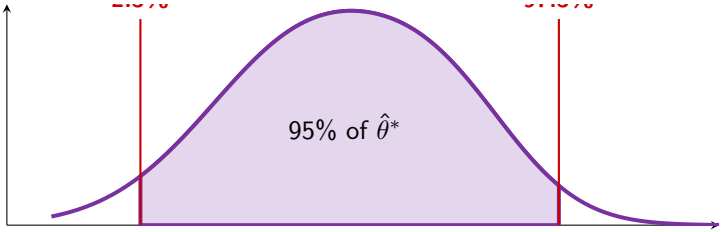

Bootstrap replicates $\hat{\theta}^*$

Advantages: respects asymmetry, no normality assumption, transformation-invariant.

Method 3: BCa (Bias-Corrected and Accelerated)

The percentile method can have poor coverage when there is **bias** or **skewness**. The BCa interval adjusts the percentile cutoffs:

$$\text{CI}_{\text{BCa}} = \left[\hat{\theta}^*_{(\alpha_1)}, \quad \hat{\theta}^*_{(\alpha_2)}\right]$$

where α_1, α_2 are **adjusted** percentiles that correct for:

$\hat{z}_0 =$ **bias correction** (median bias of bootstrap dist.)

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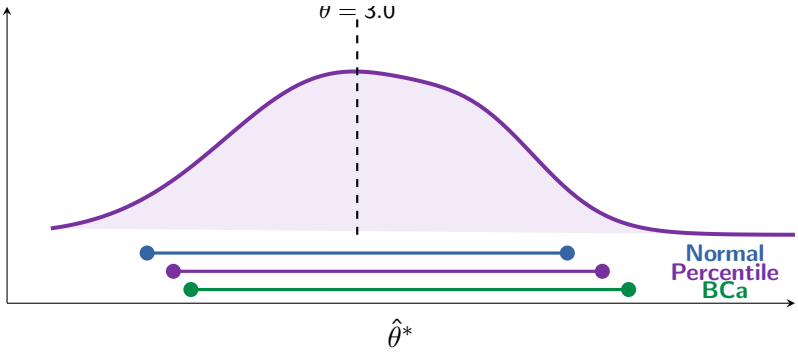
Normal: symmetric $\xrightarrow{\text{better}}$ Percentile: respects shape $\xrightarrow{\text{better}}$ BCa: corrects coverage

BCa is the **gold standard** for bootstrap CIs. Use it by default. Needs $B \geq 2{,}000$.

Comparing Bootstrap CI Methods



When the bootstrap distribution is **skewed**, the three methods give different intervals. The Normal CI is symmetric around $\hat{\theta}$ and can miss the tail. BCa adapts to the shape.

Parametric vs Nonparametric Bootstrap

Two flavors of resampling

Two Flavors of Bootstrap

Nonparametric

Resample **with replacement** from the original data

No distributional assumptions

$$x_1^*, \ldots, x_n^* \sim \hat{F}_n$$

Default choice

Parametric

Fit a model, then simulate **new data** from the fit

Assumes a distribution (e.g., Normal)

$$x_1^*, \ldots, x_n^* \sim f(\cdot \mid \hat{\theta})$$

More efficient if model is correct

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Rule of thumb: Use nonparametric unless you have strong reasons to trust the model.

Example: Bootstrap SE of the Correlation

Data: (x_i, y_i) for $i = 1, \ldots, n$. Statistic: $\hat{r} = \mathrm{Cor}(x, y) = 0.68$.

There's a formula for SE (Fisher's z-transform), but the bootstrap gives it for free:

1. Resample **pairs** (x_i^*, y_i^*) with replacement (keep pairs together!)
2. Compute $\hat{r}^* = \mathrm{Cor}(x^*, y^*)$ on each bootstrap sample
3. Repeat $B = 5{,}000$ times
4. $\mathrm{SE}_{\mathrm{boot}}(\hat{r}) = \mathrm{SD}(\hat{r}_1^*, \ldots, \hat{r}_B^*) \approx 0.09$
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Important: Resample the *pairs*, not x and y separately! Resampling them independently would destroy the correlation structure.

When Bootstrap Fails

It's not magic — it has limits

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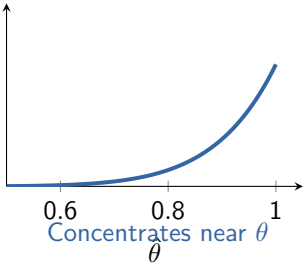
4. Non-smooth statistics: Statistics that change discontinuously (like the mode or the number of clusters) can produce poor bootstrap distributions.

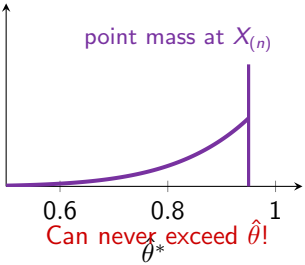
Bootstrap Fails for Extremes: Why?

Example: $X_1, \ldots, X_n \sim \text{Uniform}(0, \theta)$. Estimate $\hat{\theta} = X_{(n)} = \max(X_i)$.

True dist. of $X_{(n)}$



Bootstrap dist. of $X^*_{(n)}$



The bootstrap **underestimates** the variability of $X_{(n)}$ because it can't generate values beyond the observed maximum. Same issue from Lecture 5 (MLE for Uniform).

Permutation Tests

Resampling for hypothesis testing

The Permutation Test Idea

Question: Is there a real difference between Group A and Group B?

Step 1: Compute the test statistic: $T_{\text{obs}} = \bar{X}_A - \bar{X}_B$.

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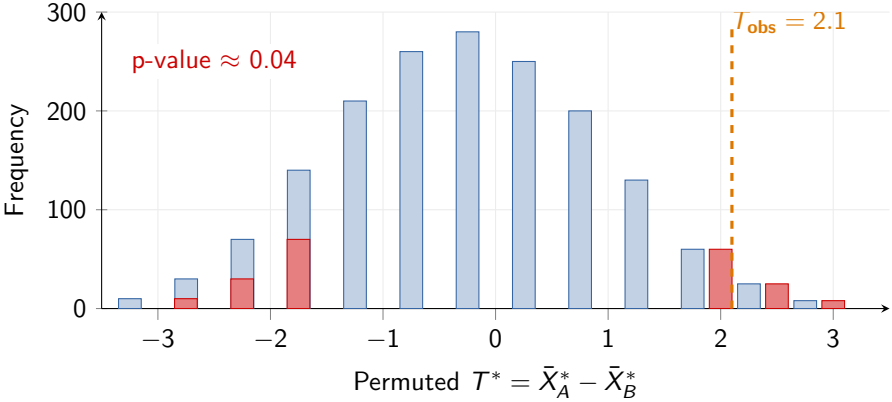
Step 5: p-value $= \frac{\#\{|T^*| \geq |T_{\text{obs}}|\}}{B}$. If small $\Rightarrow$ reject H_0.

Permutation Test: Visual Example

Drug trial: Treatment ($n_A = 8$, mean $= 5.2$) vs Control ($n_B = 8$, mean $= 3.1$). $T_{\text{obs}} = 2.1$.



Only ∼4% of shuffled datasets produce a difference ≥ 2.1. The drug effect is likely real.

Bootstrap vs Permutation: What's the Difference?

Bootstrap

Resample **with replacement** from the *same* group

Goal: estimate **SE and CI** for any statistic

Does not test a hypothesis — quantifies *uncertainty*

Permutation

Shuffle labels between groups (without replacement)

Goal: **test** H_0 that groups come from the same distribution

Generates the *null* distribution — tests whether an effect is real

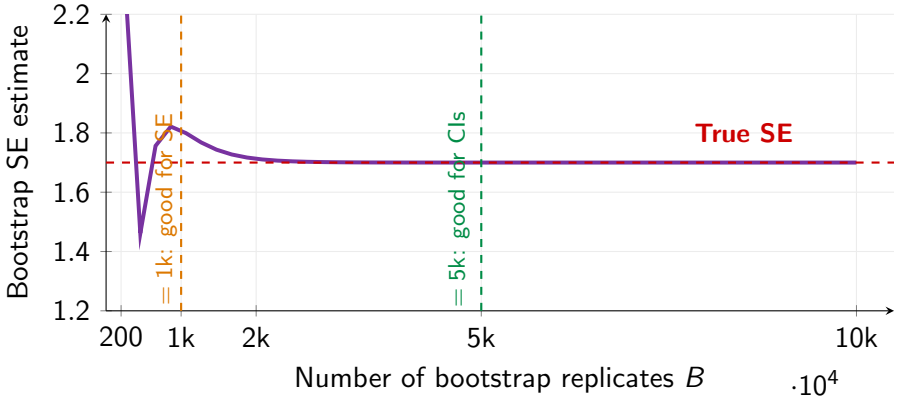
Bootstrap answers: "How precise is my estimate?"

Permutation answers: "Is there an effect?"

Practical Considerations

Making it work in practice

How Many Bootstrap Replicates?



Rules of thumb: $B \geq 200$ for SE, $B \geq 1{,}000$ for percentile CI, $B \geq 5{,}000$ for BCa CI.

Bootstrap in Python: It's Simple

```python
import numpy as np
from scipy import stats

# Original data
x = np.array([2, 5, 7, 3, 8, 1, 6])
n = len(x)

# Bootstrap
B = 10000
boot_medians = np.array([
  np.median(np.random.choice(x, size=n, replace=True))
  for _ in range(B)
])

# SE and CI
se_boot = np.std(boot_medians, ddof=1)
ci = np.percentile(boot_medians, [2.5, 97.5])
```

For BCa: use `scipy.stats.bootstrap()` (since SciPy 1.7) or `arch.bootstrap`.

Summary: The Bootstrap Toolbox

Bootstrap idea: Resample with replacement from the data to estimate the sampling distribution.

Bootstrap SE: $\mathrm{SD}(\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*)$. No formula needed — just simulation.

Bootstrap CIs: Normal (simple), Percentile (respects asymmetry), BCa (best coverage).

Parametric vs nonparametric: Use nonparametric unless you trust the model.

When it fails: Extremes, tiny n, dependent data, non-smooth statistics.

Permutation test: Shuffle labels to test H_0 of no difference. Gives exact p-values.

Bootstrap vs permutation: Bootstrap = uncertainty (SE, CI). Permutation = hypothesis test.

Practical: $B \geq 1{,}000$ for CIs. Resample *pairs* for correlated data. Use BCa by default.

Practical: Bootstrap in Action

1. **Bootstrap CI for the median:**
   - Generate $n = 30$ observations from $\text{Exp}(\lambda = 1)$
   - Compute bootstrap SE and 95% CIs (Normal, Percentile, BCa)
   - Compare: which CIs contain the true median $\ln 2 \approx 0.693$?
   - Repeat 500 times — which method has the best **coverage**?
2. **Permutation test:**
   - Generate $n_A = 20$ from $N(0.5, 1)$ and $n_B = 20$ from $N(0, 1)$
   - Permutation test with $B = 5{,}000$ shuffles
   - Compare the permutation p-value with the t-test p-value
3. **Bootstrap for correlation:**
   - Use real data (e.g., `mpg` vs `weight` from a car dataset)
   - Bootstrap the correlation; plot the bootstrap distribution
   - Is it symmetric? Compare Normal CI with BCa CI

Homework

1. The heights (cm) of 12 students are: 165, 170, 168, 172, 175, 180, 163, 178, 169, 171, 167, 174.
   (a) Compute the bootstrap SE of the median using $B = 5{,}000$.
   (b) Construct 95% Normal and Percentile bootstrap CIs.
   (c) Is the bootstrap distribution symmetric? Why or why not?
2. Show that the bootstrap estimate of $\mathrm{SE}(\bar{X})$ converges to $S/\sqrt{n}$ as $B \to \infty$.
   Hint: What is the variance of sampling with replacement from a finite set?
3. A/B test: Group A (50 users, mean revenue \$12.50, SD \$8.00) vs Group B (50 users, mean \$14.20, SD \$9.50).
   (a) Perform a permutation test for H_0: no difference in means.
   (b) Compute a bootstrap CI for the *difference* in means.
   (c) Do the two approaches agree?
4. Why does the bootstrap fail for $\hat{\theta} = X_{(n)}$ when $X_i \sim \mathrm{Uniform}(0, \theta)$?
   Show that the bootstrap distribution of $X^*_{(n)}$ has a point mass at $X_{(n)}$ of size $(1 - 1/n)^n \to 1/e$.

Recommended Visualizations & Resources

Interactive: Seeing Theory — Bootstrap (Brown University)

`seeing-theory.brown.edu/frequentist-inference` — bootstrap resampling animated step by step. Watch the bootstrap distribution build up in real time.

Video: StatQuest — Bootstrapping

`statquest.org/bootstrapping` — clearest visual explanation of the bootstrap idea. Includes worked examples and the connection to confidence intervals.

Interactive: Bootstrap Confidence Intervals (Yihui Xie)

`datascienceplus.com/bootstrap-confidence-intervals` — compare Normal, Percentile, and BCa intervals visually. See how skewness affects coverage.

Reading: Efron & Tibshirani, "An Introduction to the Bootstrap"

The foundational textbook. Chapters 1–6 cover everything in this lecture. Chapter 14 covers permutation tests.

Python: scipy.stats.bootstrap()

`docs.scipy.org/doc/scipy/reference/generated/scipy.stats.bootstrap.html` — BCa intervals built in since SciPy 1.7. Three lines of code.

Questions?

Next: Lecture 9 — Hypothesis testing, p-values, and power